PORTFOLIO PROJECT | PHASE B RESULTS

What Pushes the Monopile, and How Hard?

Wave loading and wind-vortex check for each installation stage

Project	Vibration and fatigue estimation of a bare offshore monopile before commissioning (self-study)
This phase	Generate JONSWAP sea states, compute linearised Morison wave forces on the pile, and check vortex-induced vibration (VIV) against the Phase A frequencies
Tools	Python 3 (NumPy, SciPy, Matplotlib); scripts: fem.py, phase_b_loading.py
Status	Phase B complete. All sea states, wind statistics and damping values are illustrative, not site data.

1. The idea in simple words

Phase A told us how fast each stage wobbles. Phase B asks what pushes it. Two pushers matter before the turbine is connected: **waves**, which shove the submerged pile back and forth every 6 to 12 seconds, and **wind**, which can shed swirls (vortices) behind the tower at a steady rhythm. Like pushing a swing, a push only builds up motion if its rhythm matches the swing's own. So for every stage we compare the pushing rhythm with the natural frequency from Phase A.

2. Key findings at a glance

#	Finding	In simple words
1	Waves push the pile by inertia, not by drag	The pile is wide (8 m), so waves mostly accelerate the water around it. Drag is below 0.5 % of the moment variance in all six sea states (KC at most 3.3). Hs = 4 m, Tp = 8 s gives a mudline moment of about 19 MNm RMS and about 74 MNm expected 3-hour maximum.
2	Bare pile and pile + transition piece follow waves almost statically	Dynamic amplification is 1.02 and 1.03. Their frequencies (about 0.6 to 1.5 Hz) are far above the wave band, so waves are not the fatigue driver here.
3	Tower without rotor: waves are fine, wind is the risk	Wave amplification is only 1.07 to 1.22. But the top section locks in to the wind at 9 to 13 m/s, a normal wind speed. With the illustrative wind statistics the tower sits inside the +/-20 % lock-in band about 28 to 30 % of the time.
4	Phase A hypothesis confirmed, with one gap closed	Phase A flagged the tower without rotor. Phase B confirms it and also shows that the Phase A VIV table left out the full turbine: its top section gives 5.0 to 6.1 m/s. Phase C must decide how much damage the VIV exposure actually causes.

3. What was modelled

Part	What it is	Key values (illustrative)
Structure	Phase A beam model, rebuilt from the Phase A-2 parameter table	First frequencies 1.22 / 0.86 / 0.51 / 0.26 Hz (medium soil), same as Phase A. Mode 1, 2, 3 of the bare pile: 1.22 / 7.57 / 20.38 Hz.
Sea states	JONSWAP, peak factor from the DNV rule, normalised to exact Hs	Six cases, Hs 1 to 4 m, Tp 6 to 12 s; steepness Hs/Lp between 1/63 and 1/25
Wave kinematics	Linear (Airy) waves, 30 m depth, exact dispersion relation	Loaded from the seabed to still-water level; no wave stretching
Wave force	Morison: inertia Cm = 2.0 plus drag Cd = 1.0, drag linearised (Borgman)	Inertia corrected for diffraction (MacCamy-Fuchs), see section 5
Wind VIV	Strouhal relation, St = 0.2; Weibull wind, power-law shear	A = 11 m/s, k = 2.1 at hub height 140 m, shear exponent 0.11; lock-in band 0.8 to 1.2 x U_crit

4. Check: is the Phase A model still correct?

The Phase A script was not part of this phase's inputs, so the beam model was rebuilt from the Phase A-2 parameter table. It reproduces the bare-pile modes (1.22 / 7.57 / 20.38 Hz) and all four first frequencies, as well as the soft, medium, stiff soil results and the no-water results, to two decimals (one rounding difference: soft soil bare pile 0.82 vs 0.83 Hz). The Phase A *rigid base* column was not reproduced (bare pile 3.46 vs 3.66 Hz), probably a different fixity convention. It is not used in Phase B.

5. Result 1: the sea states and the force on the pile

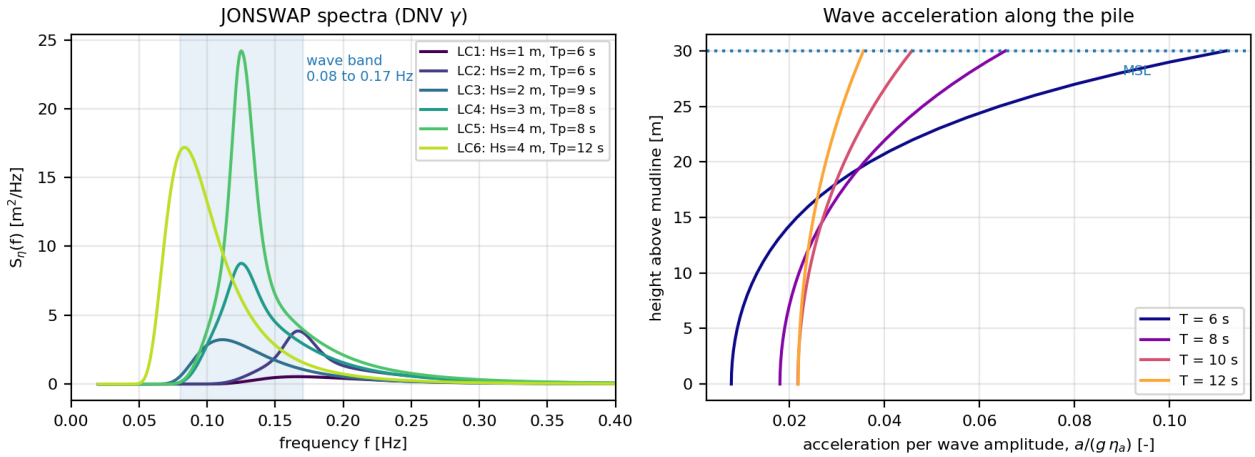


Figure 1: JONSWAP spectra (left) and wave acceleration along the pile (right). Short waves shake only the top of the pile; long waves reach the seabed.

Linearised Morison. The inertia force follows the wave acceleration; the drag force follows the velocity, which is 90 degrees out of phase, so the two variances simply add. Drag is linearised with the sea state's own velocity RMS at each height (Borgman).

Diffraction correction. Morison's equation is valid for D/L below about 0.2. For the 8 m pile this fails for short waves: D/L is already 0.14 at T = 6 s, and the 0.26 Hz resonance of the full turbine corresponds to a 24 m wavelength (D/L about 0.34). The inertia force is therefore multiplied by the MacCamy-Fuchs ratio (1.01 at 0.17 Hz, 0.63 at 0.26 Hz, 0.10 at 0.5 Hz). Without it, the top motion of the tower stages would be overestimated by about 40 to 70 %.

Case	Hs [m]	Tp [s]	gamma	Tz [s]	Base shear RMS [MN]	Mudline moment RMS [MNm]	3 h max moment [MNm]	KC	Drag share
LC1	1	6	1.0	4.3	0.22	5.0	20	0.8	0.0 %
LC2	2	6	2.4	4.6	0.46	10.2	40	1.6	0.0 %
LC3	2	9	1.0	6.4	0.47	9.3	36	1.6	0.0 %
LC4	3	8	1.6	5.9	0.72	14.5	56	2.3	0.1 %
LC5	4	8	3.2	6.2	0.97	19.2	74	3.1	0.2 %
LC6	4	12	1.0	8.5	0.88	15.8	60	3.3	0.3 %

Wave loading on the pile, with diffraction correction. The loads are identical for all four stages because only the pile is in the water. Expected maximum assumes a Rayleigh distribution over 3 hours.

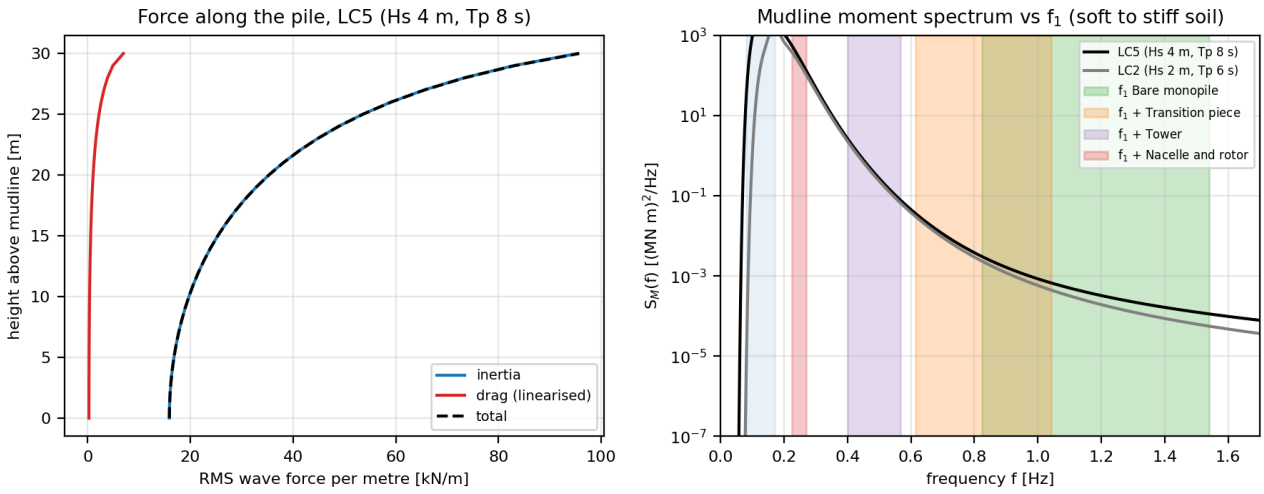


Figure 2: Force per metre along the pile (left) and mudline moment spectrum against the first frequencies of each stage, soft to stiff soil (right). The shaded blue band is the wave band.

6. Result 2: how much do the stages amplify the waves?

To compare stages fairly, the wave force was projected onto mode 1 of each stage and the response of a single-degree-of-freedom oscillator was computed with 1 % damping (soil and hydrodynamic damping only; no aerodynamic damping because the rotor is absent or ignored). The amplification factor is the dynamic RMS response divided by the quasi-static RMS response.

Case	Bare monopile DAF	+ Transition piece DAF	+ Tower DAF	+ Nacelle and rotor DAF	Top motion RMS [mm] bare pile / + TP / + tower / + RNA
LC1	1.03	1.05	1.22	6.56	3 / 5 / 17 / 75
LC2	1.02	1.05	1.19	5.59	7 / 10 / 33 / 133
LC3	1.02	1.03	1.11	3.54	7 / 9 / 29 / 79
LC4	1.02	1.04	1.13	3.94	10 / 14 / 46 / 136
LC5	1.02	1.03	1.11	3.51	14 / 19 / 60 / 161
LC6	1.01	1.02	1.07	2.50	12 / 16 / 49 / 96

Dynamic amplification factor (DAF) with 1 % damping, medium soil, MacCamy-Fuchs corrected. Top motion is the dynamic RMS of the highest point of each stage.

Main message: only the finished turbine amplifies waves strongly (DAF 2.5 to 6.6), because its 0.26 Hz sits just above the wave band and the tail of the spectrum feeds it. The three bare stages are almost quasi-static. Waves are a modest fatigue driver for the bare stages; wind VIV is the more suspicious one.

Limits of this step. The quasi-static part uses mode 1 only, which is a rough approximation of the static deflection under a load applied at the pile. The DAF for the full turbine depends strongly on damping (resonant amplitude scales roughly with 1/damping), so it is only a sign, not a result. Phase C will compute stress properly from the full response, and Phase D will vary damping in the Monte Carlo study.

7. Result 3: the wind-vortex (VIV) check

Wind flowing around a tube sheds vortices at f_s = St x U / D. Trouble starts when f_s meets a natural frequency, at the critical wind speed $U_c = f_1$ x D / St. The check uses the top section of each stage, because mode 1 moves most there and the tapered tower is thinnest there (lowest U_c).

Stage	Diameter at top section	f1 soft / medium / stiff [Hz]	Critical wind speed U_c [m/s]	Time in lock-in band (6 months)	Verdict
Bare monopile	8.0 m at 40 m	0.82 / 1.22 / 1.54	33.0 / 48.6 / 61.6	< 0.1 %	Too high, no concern
+ Transition piece	8.4 m at 55 m	0.61 / 0.86 / 1.04	25.8 / 36.3 / 43.8	< 0.1 %	Too high, no concern
+ Tower	4.5 m at 140 m	0.40 / 0.51 / 0.57	9.0 / 11.5 / 12.7	30.3 % (1328 h)	Within normal wind: check closely
+ Nacelle and rotor	4.5 m at 140 m	0.22 / 0.26 / 0.27	5.0 / 5.8 / 6.1	16.6 % (728 h)	Within normal wind; rotor and aero damping not modelled

Critical wind speeds with St = 0.2. The time in the band uses medium soil, Weibull wind scaled to the height of the section, and a lock-in band of 0.8 to 1.2 x U_c. The band width and the wind statistics are assumptions.

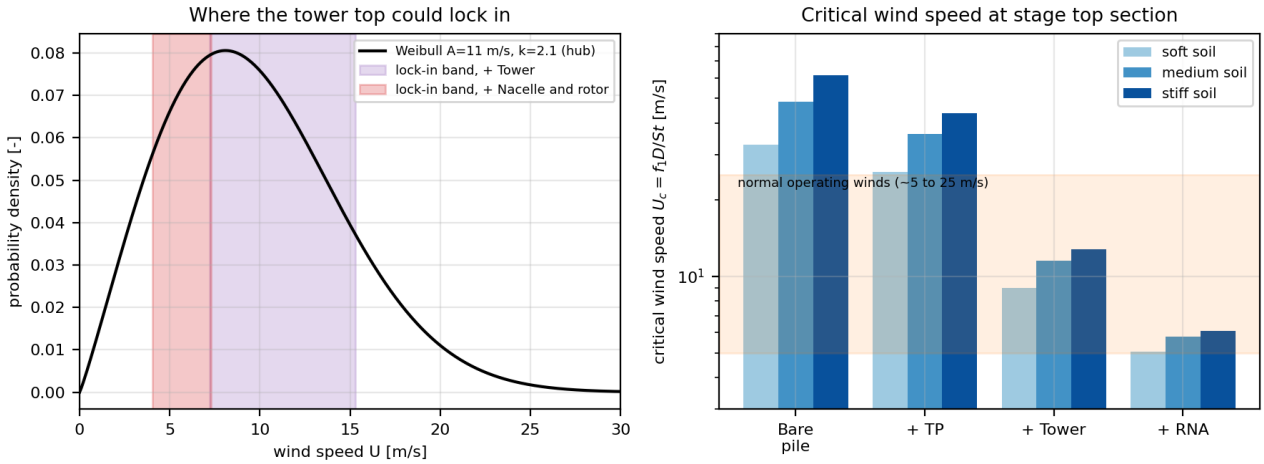


Figure 3: The lock-in bands sit on the bulk of the wind distribution for the tower stages (left). Critical wind speed per stage against the range of operating winds (right).

The Phase A hand estimate of 9 to 19 m/s for the tower without rotor is reproduced; 19 m/s belongs to the 6.5 m base section, while the top section (4.5 m) gives 9 to 12.7 m/s. Mode 2 of the tower stage gives about 32 m/s at the top, which is rare, so mode 1 governs.

Take-away: being inside the lock-in band is an *exposure*, not damage. Whether the tower actually vibrates and how large the amplitude gets depends on damping, turbulence and surface roughness. Phase C must turn this exposure into stress cycles, and the Strouhal number and lock-in range still need to be checked against a standard (for example DNV-RP-C205) or a textbook before reporting.

8. Limitations (to state in the final report)

- Linear waves, no wave stretching above still-water level, no breaking waves or slamming, no marine growth, no current or wind drag yet (optional items of the task sheet, not done).
- MacCamy-Fuchs applies to the inertia term of a smooth vertical cylinder in unidirectional linear waves; drag is unchanged.
- Wave-band energy for T = 6 s sea states extends beyond real steepness limits at high frequency, because the JONSWAP tail is a fitted shape.
- Single-mode projection and 1 % damping; no aerodynamic damping, no rotor inertia, no soil nonlinearity.
- Strouhal number, lock-in band width (+/-20 %) and the Weibull wind (A = 11 m/s, k = 2.1) are illustrative assumptions; amplitudes of VIV are not computed.
- All values are illustrative and not from a real site, so quote them to two decimals at most.

9. Next steps

- Phase C: turn the modal force spectrum $S_Q(f)$ of each stage into response, mudline stress, rainflow cycles and damage for a 6-month waiting period; add a VIV amplitude model for the tower stage as the most suspicious case.
- Phase D: vary soil stiffness, damping and wave height (and the VIV inputs) in the Monte Carlo study; use the DAF and VIV exposure to define the logger band and sampling rate.